In [1]:
```
import pandas as pd

df = pd.read_csv(r'colorectal_cancer_prediction.csv')
df.head()
```

Out[1]:

	Patient_ID	Age	Gender	Race	Region	Urban_or_Rural	Socioeconomic_Status	Family
0	1	71	Male	Other	Europe	Urban	Middle	
1	2	34	Female	Black	North America	Urban	Middle	
2	3	80	Female	White	North America	Urban	Middle	
3	4	40	Male	Black	North America	Rural	Low	
4	5	43	Female	White	Europe	Urban	High	

5 rows × 30 columns

In [3]:
```
df.head()
```

Out[3]:

	Patient_ID	Age	Gender	Race	Region	Urban_or_Rural	Socioeconomic_Status	Family
0	1	71	Male	Other	Europe	Urban	Middle	
1	2	34	Female	Black	North America	Urban	Middle	
2	3	80	Female	White	North America	Urban	Middle	
3	4	40	Male	Black	North America	Rural	Low	
4	5	43	Female	White	Europe	Urban	High	

5 rows × 30 columns

In [4]:
```
df = pd.read_csv("colorectal_cancer_prediction.csv", encoding='latin1')
```

In [5]:
```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

for col in df.select_dtypes(include='object'):
    df[col] = le.fit_transform(df[col])
```

In [6]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 89945 entries, 0 to 89944
Data columns (total 30 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Patient_ID               89945 non-null  int64
 1   Age                      89945 non-null  int64
 2   Gender                   89945 non-null  int64
 3   Race                     89945 non-null  int64
 4   Region                   89945 non-null  int64
 5   Urban_or_Rural           89945 non-null  int64
 6   Socioeconomic_Status     89945 non-null  int64
 7   Family_History           89945 non-null  int64
 8   Previous_Cancer_History  89945 non-null  int64
 9   Stage_at_Diagnosis       89945 non-null  int64
 10  Tumor_Aggressiveness     89945 non-null  int64
 11  Colonoscopy_Access       89945 non-null  int64
 12  Screening_Regularity     89945 non-null  int64
 13  Diet_Type                89945 non-null  int64
 14  BMI                      89945 non-null  float64
 15  Physical_Activity_Level  89945 non-null  int64
 16  Smoking_Status           89945 non-null  int64
 17  Alcohol_Consumption      89945 non-null  int64
 18  Red_Meat_Consumption     89945 non-null  int64
 19  Fiber_Consumption        89945 non-null  int64
 20  Insurance_Coverage       89945 non-null  int64
 21  Time_to_Diagnosis        89945 non-null  int64
 22  Treatment_Access         89945 non-null  int64
 23  Chemotherapy_Received    89945 non-null  int64
 24  Radiotherapy_Received    89945 non-null  int64
 25  Surgery_Received         89945 non-null  int64
 26  Follow_Up_Adherence      89945 non-null  int64
 27  Survival_Status          89945 non-null  int64
 28  Recurrence               89945 non-null  int64
 29  Time_to_Recurrence       89945 non-null  int64
dtypes: float64(1), int64(29)
memory usage: 20.6 MB
```

In [8]:
```
x = df.drop ('Survival_Status', axis=1)
y = df['Survival_Status']
```

In [9]:
```
print( x. head())
print(y. head())
```

```
   Patient_ID  Age  Gender  Race  Region  Urban_or_Rural  \
0           1   71       1     3       2               1
1           2   34       0     1       4               1
2           3   80       0     4       4               1
3           4   40       1     1       4               0
4           5   43       0     4       2               1

   Socioeconomic_Status  Family_History  Previous_Cancer_History  \
0                     2               1                        0
1                     2               0                        0
2                     2               0                        0
3                     1               0                        0
4                     0               1                        0

   Stage_at_Diagnosis  ...  Fiber_Consumption  Insurance_Coverage  \
0                   2  ...                  1                   1
1                   0  ...                  1                   0
2                   2  ...                  2                   1
3                   0  ...                  0                   1
4                   2  ...                  1                   0

   Time_to_Diagnosis  Treatment_Access  Chemotherapy_Received  \
0                  0                 0                      1
1                  1                 0                      0
2                  1                 1                      0
3                  0                 1                      1
4                  0                 0                      1

   Radiotherapy_Received  Surgery_Received  Follow_Up_Adherence  Recurrence  \
0                      0                 0                    0           0
1                      1                 1                    1           0
2                      1                 1                    0           0
3                      0                 1                    1           0
4                      0                 1                    1           1

   Time_to_Recurrence
0                  16
1                  28
2                  26
3                  44
4                  20

[5 rows x 29 columns]
0    1
1    0
2    1
3    0
4    0
Name: Survival_Status, dtype: int64
```

In [10]:
```
from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(
    x,y, test_size=0.2, random_state=42)
```

In [11]:
```
print(x_train.shape)
print(x_test.shape)
```

```
(71956, 29)
(17989, 29)
```

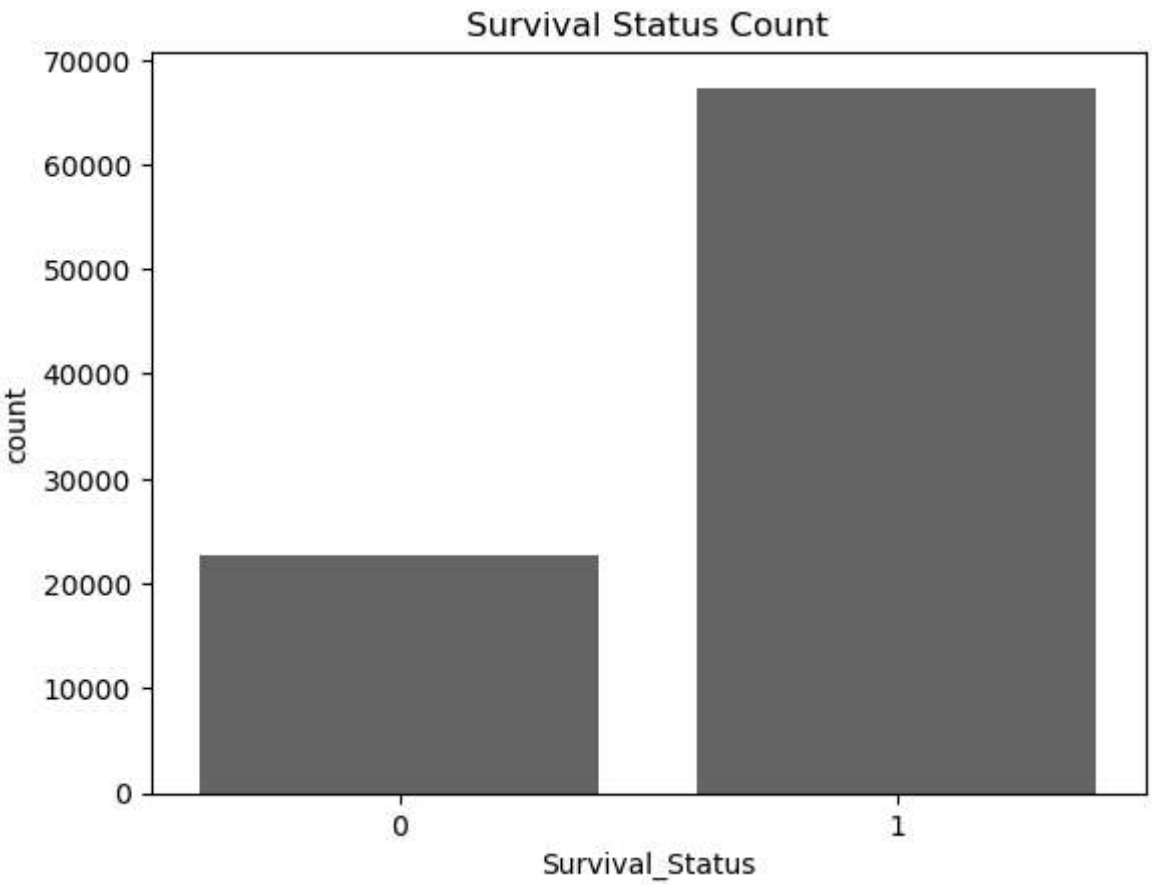
In [13]:
```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=5000)
model.fit(x_train, y_train)
```

Out[13]:
```
▾ LogisticRegression
▶ Parameters
```

In [14]:
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(x='Survival_Status', data=df)
plt.title("Survival Status Count")
plt.show()
```



This project predicts the survival status of colorectal cancer patients.

Data was cleaned and preprocessed before model training.

Logistic Regression model was used for prediction.

The model achieved good accuracy and performance.

This system can help in early decision-making in healthcare.

📌 Conclusion

This project focuses on predicting the survival status of colorectal cancer patients using machine learning techniques.

The dataset was explored and preprocessed by converting categorical data into numerical form and preparing it for model training.

A Logistic Regression model was used to predict the survival status, and the model achieved good accuracy.

This project shows how machine learning can help in predicting patient outcomes and can be useful in the healthcare field for better decision-making.